

Data Collection and Preprocessing Phase

Date	17 August 2026
Team ID	XXXXXX
Project Title	India's Agricultural crop production Analysis
Maximum Marks	10 Marks

Data Exploration and Preprocessing Template

Identifies data sources, assesses quality issues like missing values and duplicates, and implements resolution plans to ensure accurate and reliable analysis.

Section	Description
Data Overview	• Schema Inspection: Examined dataset attributes including, Season, Crop, Area (Hectares), and Production. • Dataset Profiling: Assessed missing values, structural irregularities, and value distributions across 30+ states, 600+ districts, and 50+ crop varieties from 1997 to 2021.
Data Cleaning	Data Cleaning Handled null values in production through localized median yield estimation; dropped corrupted records with invalid coordinates or zero-area entries. Removed duplicate rows and noise caused by legacy manual data entry.
Data Transformation	Calculated the standardized productivity metric $\text{Yield (Tones/Hectare)} = \frac{\text{Production}}{\text{Area}}.$ Normalization: Grouped low-frequency minor crops into broad categories (e.g., Pulses, Oilseeds, Cereals) for macro-level trend analysis.

Data Type Conversion	Data Type Conversion Cast Area and Production fields to 64-bit floating point for aggregation accuracy; converted Crop Year to discrete temporal values and encoded categorical dimensions into structured string objects.
Column Splitting and Merging	Standardization: Cleaned and merged inconsistent state/district naming variations resulting from geographical boundary redefinitions over time. Attribute Parsing: Split combined seasonal text attributes (e.g., combining climate tags) into standardized season labels (Kharif, Rabi, Summer, Whole Year).
Data Modeling	Data Modeling Designed an optimized Star Schema with a centralized fact table (Crop Production Fact) linked to lookup dimensions (Time Dimension, Geography Dimension, Crop Dimension). rapid Tableau query execution
Save Processed Data	? Storage Format: Exported clean data to optimized .csv /. parquet formats and pushed structured tables to SQL database staging tables. ? Tableau Extraction: Created a hyper-extracted data source file (. hyper) ready for rapid visualization render in